

# 用戶畫像探索專案： 基於 NMF 的影視分群分析

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01

# 專案動機與設計

資料來源: MovieLens Latest Small Dataset (2018)

# 專案動機

- The Challenge: 原始數據中我們只知道「誰給了幾分」，但不知道「為什麼」
- The Goal: 從雜亂的評分數據中，解碼出 "Hidden Content Genes" 與 "User Taste Groups"

# 分析方法



Raw Data

movies.csv  
ratings.csv



NMF

Deconstruct movies  
into 6 latent features.



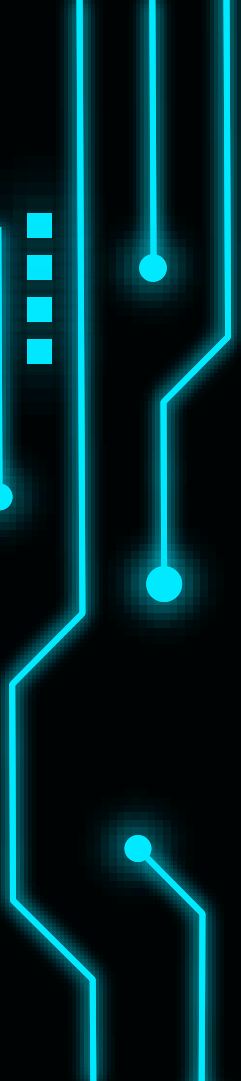
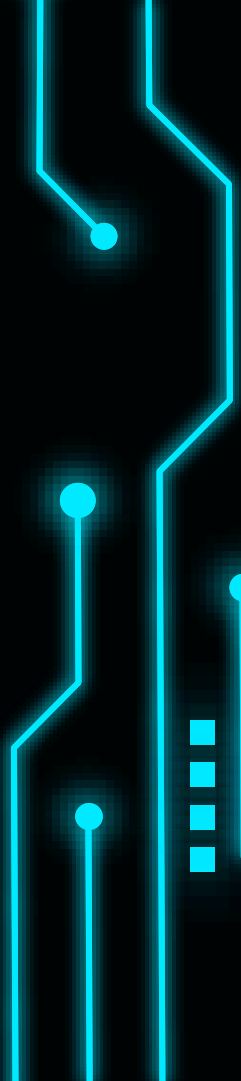
K-Means

Cluster users into 4  
segments.



PCA

Map high-dimensional  
data to 2D market map.



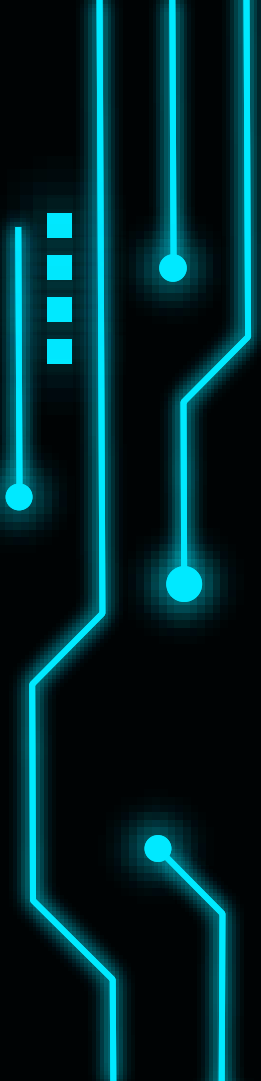
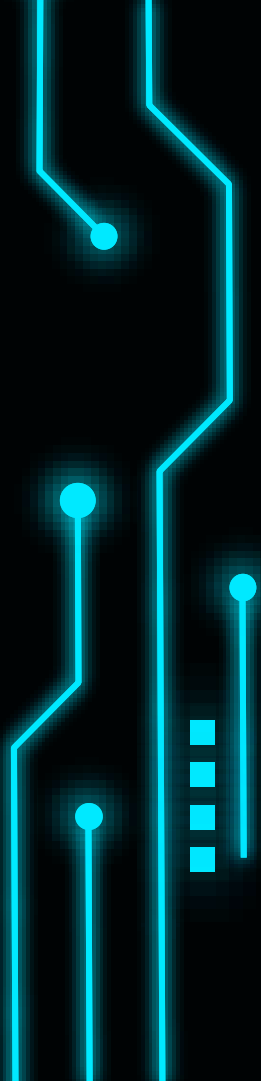
02

# 核心發現一

Hidden Content Genes

# Hidden Content Genes

|                              |                                                                               |
|------------------------------|-------------------------------------------------------------------------------|
| <b>2000s Animated Family</b> | Shrek (2001) Finding Nemo (2003) Monsters, Inc. (2001)                        |
| <b>90s Epic Classics</b>     | Forrest Gump (1994) Apollo 13 (1995) The Fugitive(1993)                       |
| <b>Modern CGI</b>            | The Dark Knight (2008) Inception (2010) Iron Man (2008)                       |
| <b>Cult Classics</b>         | Fight Club (1999) Pulp Fiction (1994)<br>The Shawshank Redemption (1994)      |
| <b>80s Sci-Fi/Action</b>     | Star Wars: Episode V - The Empire Strikes Back (1980)<br>The Terminator(1984) |
| <b>Old School</b>            | Casablanca (1942) Rear Window (1954) Psycho (1960)                            |



03

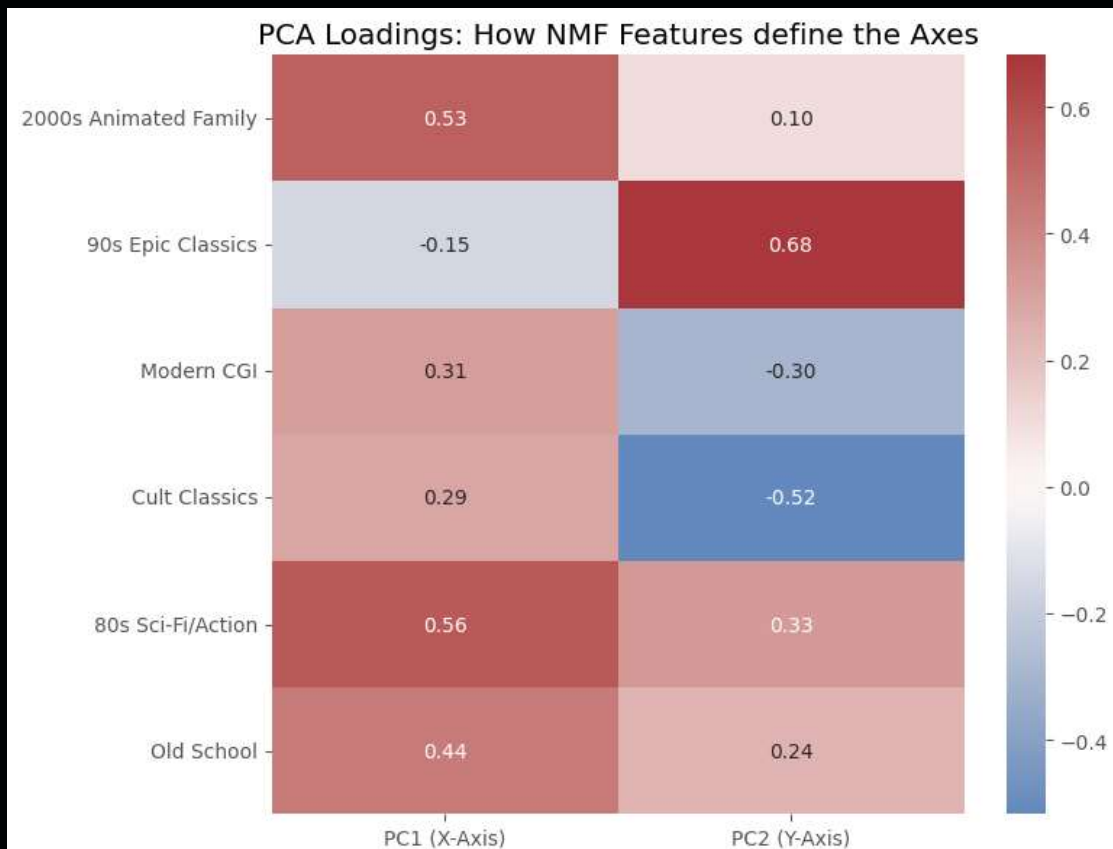
## 核心發現二

User Taste Groups

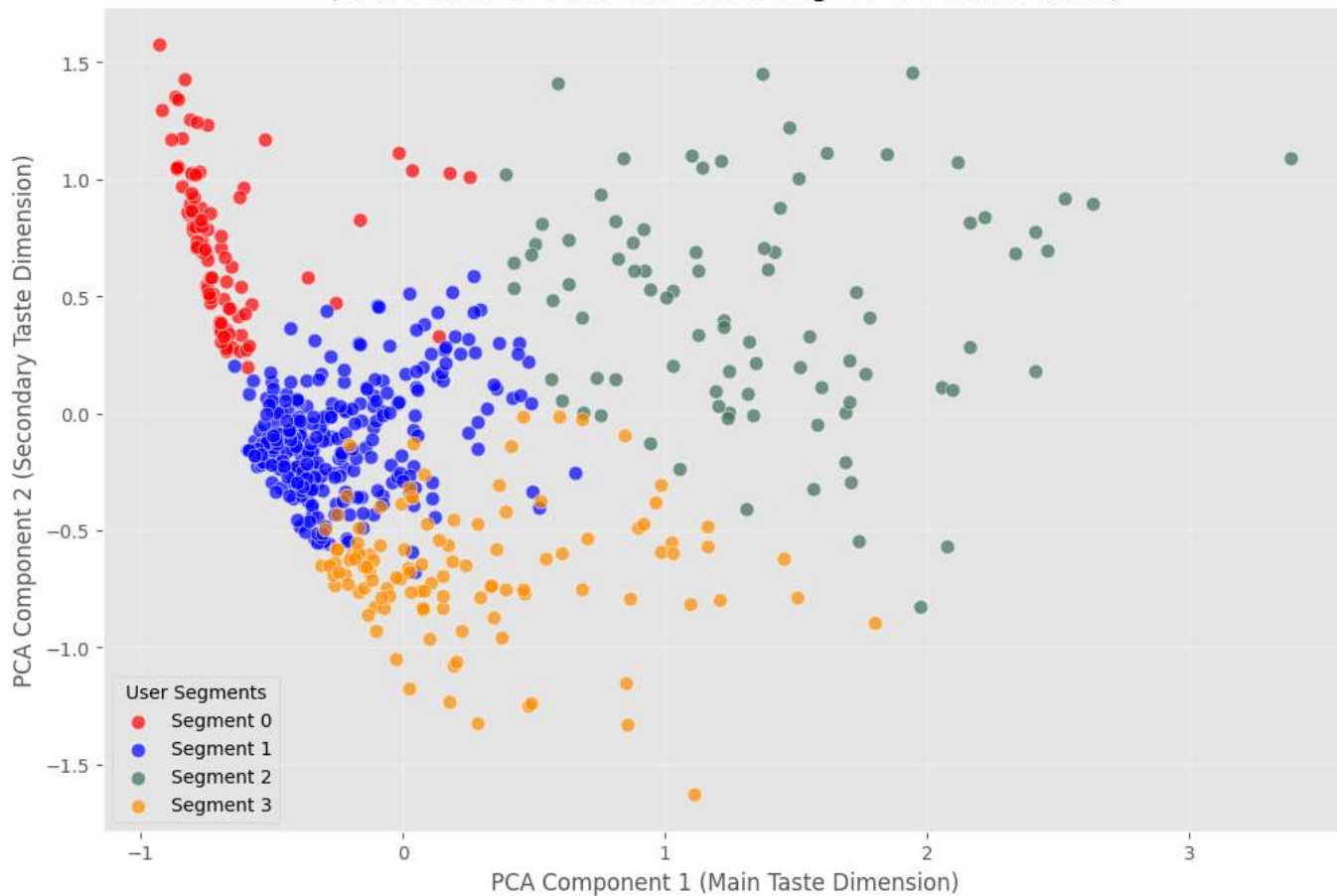


# PC1(X-Axis) 大眾娛樂程度

## PC2(Y-Axis) 電影風格光譜

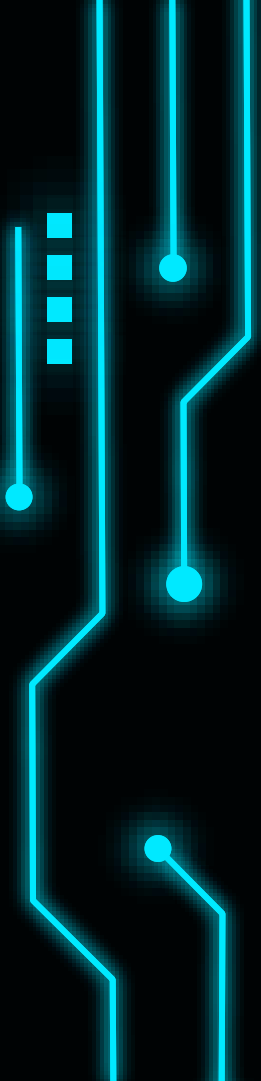
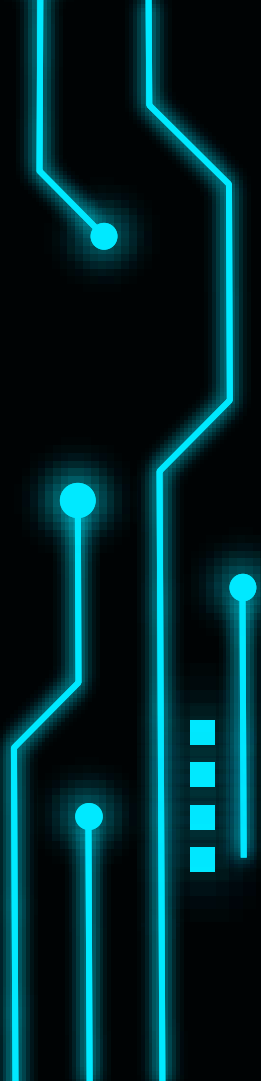


User Segmentation Map  
(NMF Features + KMeans Clustering + PCA Visualization)



# User Segmentation Profile

| Cluster ID | Top Motivation    | Secondary Motivation  |
|------------|-------------------|-----------------------|
| Cluster 0  | 90s Epic Classics | 80s Sci-Fi/Action     |
| Cluster 1  | 80s Sci-Fi/Action | Cult Classics         |
| Cluster 2  | 80s Sci-Fi/Action | 2000s Animated Family |
| Cluster 3  | Cult Classics     | Modern CGI            |



04

# 商業洞察

客戶分析及行銷策略

### Cluster 0 (The Nostalgic)

- 偏好 90 年代經典劇情片
- 建立「奧斯卡金像獎」或「影史經典」專區，推廣兼具內容深度與質感的電影

### Cluster 1 (The Casual)

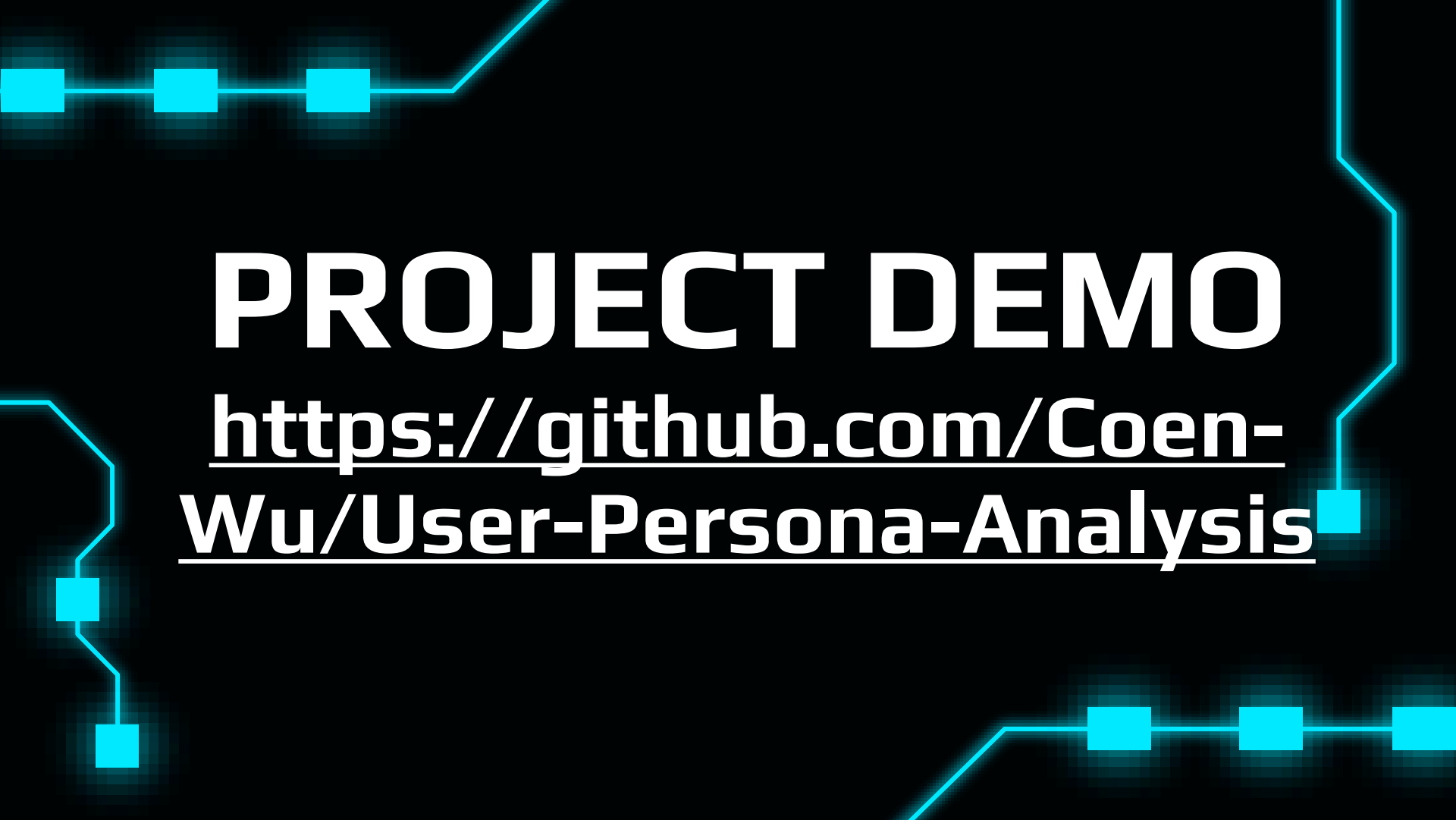
- 所有特徵分數都偏低，缺乏明確喜好，流失風險最高
- 主動推播「今日 Top 10」熱門榜單，減少他們的選擇障礙，避免退訂

### Cluster 2 (The Geeks)

- 在科幻與動畫的得分極高，喜歡流行電影
- 推廣周邊以及提供「新片搶先看」的付費特權

### Cluster 3 (The Edgy)

- 喜愛暗黑風格與非主流題材
- 避免推薦大眾喜劇，改成推薦冷門神作，建立平台「懂我不俗」的品牌認同



# PROJECT DEMO

<https://github.com/Coen-Wu/User-Persona-Analysis>